

A several-variable growing-weight $\bar{\partial}$ estimate

Candidate full-result packet for arXiv:1311.2020

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Outcome. Hedenmalm's Theorem 1.2 extends to every $\mathbb{C}^n$ with the same factor 1/2. The Laplacian becomes the trace of the Levi form, and the holomorphic-moment condition becomes annihilation against anti-holomorphic-divergence-free coefficient vectors. Positivity of the Levi trace everywhere suffices; no uniform lower bound is required.

1. The exact question and answer

For $z = (z_1, \dots, z_n)$ write

$$\phi_{j\bar{j}} = \partial_j \bar{\partial}_j \phi, \quad L_\phi = \sum_{j=1}^n \phi_{j\bar{j}}.$$

The source proves the following on $\mathbb{C}$: if $\Delta\phi > 0$ and $f \in L^2(e^{2\phi})$ annihilates every entire $g \in L^2(e^{-2\phi})$, then $\bar{\partial}u = f$ and

$$\int |u|^2 e^{2\phi} \Delta\phi \leq \frac{1}{2} \int |f|^2 e^{2\phi}.$$

Remark 1.3 asks for a several-variable generalization.

The corresponding norms are

$$\|g\|_{L^2(\mathbb{C}, e^{-2\phi})}^2 = \int_{\mathbb{C}} |g|^2 e^{-2\phi} dA, \quad \|g\|_{L^2(\mathbb{C}, e^{2\phi})}^2 = \int_{\mathbb{C}} |g|^2 e^{2\phi} dA.$$

The space of entire functions in $L^2(\mathbb{C}, e^{-2\phi})$ is denoted by $A^2(\mathbb{C}, e^{-2\phi})$. We recall the standing assumption that ϕ be C^2 -smooth with $\Delta\phi > 0$ everywhere.

Theorem 1.2. Suppose $f \in L^2(\mathbb{C}, e^{2\phi})$ meets the condition [\(1.2\)](#) for all $g \in A^2(\mathbb{C}, e^{-2\phi})$. Then there exists a solution to the $\bar{\partial}$ -equation $\bar{\partial}u = f$ with

$$\int_{\mathbb{C}} |u|^2 e^{2\phi} \Delta\phi dA \leq \frac{1}{2} \int_{\mathbb{C}} |f|^2 e^{2\phi} dA.$$

Superficially, this theorem looks quite different than Hörmander's Theorem [1.1](#). However, it is in a sense which can be made precise dual to Theorem [1.1](#). In our presentation, we will derive it from the same rather elementary calculation which Hörmander uses in e.g. [\[10\]](#), p. 250. As such, Theorem [1.2](#) may well be known, but we have not found a specific reference.

We might add that if the polynomials are dense in the space $A^2(\mathbb{C}, e^{-2\phi})$ of entire functions, as they are, e.g., for radial ϕ , then the condition [\(1.2\)](#) needs to be verified only for monomials $g(z) = z^j$, $j = 0, 1, 2, \dots$

Remark 1.3. Naturally, Theorem [1.2](#) should generalize to the setting of several complex variables.

Figure 1: Theorem 1.2 and the several-variable question, source PDF page 2.

Define $\mathcal{Z}_\phi$ to be the coefficient vectors $g = (g_1, \dots, g_n)$ such that

$$\int_{\mathbb{C}^n} |g|^2 e^{-2\phi} dV < \infty, \quad \sum_{j=1}^n \bar{\partial}_j g_j = 0 \quad \text{in distributions.}$$

Theorem 1 (Growing weights on $\mathbb{C}^n$). *Let $\phi \in C^2(\mathbb{C}^n; \mathbb{R})$ satisfy $L_\phi > 0$ everywhere. Let $f = \sum_{j=1}^n f_j d\bar{z}_j$ obey*

$$\int_{\mathbb{C}^n} |f|^2 e^{2\phi} dV < \infty, \quad \int_{\mathbb{C}^n} \sum_{j=1}^n f_j g_j dV = 0 \quad (g \in \mathcal{Z}_\phi). \quad (\text{C})$$

Then there is $u \in L_{\text{loc}}^2(\mathbb{C}^n)$ with $\bar{\partial}u = f$ in distributions and

$$\int_{\mathbb{C}^n} |u|^2 e^{2\phi} L_\phi dV \leq \frac{1}{2} \int_{\mathbb{C}^n} |f|^2 e^{2\phi} dV. \quad (1)$$

For $n = 1$, this is exactly Theorem 1.2 of the source.

Proof intuition. Conjugate the equation by the growing weight. The resulting operator T has two decisive properties. First, the difference of two squares in the source's one-variable identity sums over the coordinates and controls a scalar by $2L_\phi$. Second, the kernel of T^* becomes exactly $\mathcal{Z}_\phi$ after undoing the weight and taking a conjugate. Condition (C) therefore puts the datum in the closure of the range of T ; coercivity turns a range-approximating sequence into an actual locally convergent solution.

2. The conjugated identity

On $C_c^\infty(\mathbb{C}^n)$ define the vector-valued operator

$$Tv = (T_1 v, \dots, T_n v), \quad T_j v = \bar{\partial}_j v - v \bar{\partial}_j \phi = e^\phi \bar{\partial}_j (e^{-\phi} v).$$

Lemma 1 (Trace-Levi coercivity). *For every $v \in C_c^\infty(\mathbb{C}^n)$,*

$$\|Tv\|_2^2 - \sum_{j=1}^n \|\partial_j v + v \partial_j \phi\|_2^2 = 2 \int_{\mathbb{C}^n} |v|^2 L_\phi dV, \quad (2)$$

$$2 \int_{\mathbb{C}^n} |v|^2 L_\phi dV \leq \|Tv\|_2^2. \quad (3)$$

Proof. For a fixed j , integration by parts in z_j gives the source's identity

$$\|\bar{\partial}_j v - v \bar{\partial}_j \phi\|_2^2 - \|\partial_j v + v \partial_j \phi\|_2^2 = 2 \int_{\mathbb{C}^n} |v|^2 \phi_{j\bar{j}} dV.$$

The remaining variables are merely parameters. Sum over j to obtain (2); discard the nonnegative square to obtain (3). $\square$

3. Compatibility is the adjoint kernel

Let $H_0 = L^2(\mathbb{C}^n)$ and $H_1 = L^2(\mathbb{C}^n; \mathbb{C}^n)$, both unweighted, and regard $T_0 = T|_{C_c^\infty}$ as a densely defined map $H_0 \rightarrow H_1$. Distributionally,

$$T_0^* k = - \sum_{j=1}^n (\partial_j + \partial_j \phi) k_j. \quad (4)$$

If $k \in H_1$ and $g_j = e^\phi \overline{k_j}$, then

$$\int |g|^2 e^{-2\phi} dV = \|k\|_2^2, \quad \overline{T_0^* k} = -e^{-\phi} \sum_{j=1}^n \bar{\partial}_j g_j.$$

Thus $k \mapsto g$ is a bijection from $\ker T_0^*$ onto $\mathcal{Z}_\phi$.

Put $h = e^\phi(f_1, \dots, f_n) \in H_1$. With the standard complex Hilbert pairing, for the corresponding g we have

$$\langle h, k \rangle_{H_1} = \int_{\mathbb{C}^n} \sum_{j=1}^n f_j g_j dV.$$

Condition (C) is consequently equivalent to $h \perp \ker T_0^*$. The elementary Hilbert-space identity

$$(\ker T_0^*)^\perp = \overline{\text{ran } T_0}^{H_1} \quad (5)$$

shows that there are $v_m \in C_c^\infty(\mathbb{C}^n)$ with $T_0 v_m \rightarrow h$ in H_1 .

4. Closure produces the solution

Apply (3) to $v_m - v_\ell$. Then

$$\int |v_m - v_\ell|^2 L_\phi \, dV \leq \frac{1}{2} \|T_0 v_m - T_0 v_\ell\|_2^2,$$

so (v_m) converges to some v in $L^2(L_\phi \, dV)$ and

$$\int |v|^2 L_\phi \, dV \leq \frac{1}{2} \|h\|_2^2. \quad (6)$$

Because L_ϕ is continuous and strictly positive, it has a positive minimum on each compact set. Hence $v_m \rightarrow v$ also in L_{loc}^2 . Passing to distributions in $T_0 v_m \rightarrow h$ gives $Tv = h$ exactly.

Set $u = e^{-\phi} v$. The intertwining formula defining T yields

$$e^\phi \bar{\partial} u = Tv = h = e^\phi f,$$

so $\bar{\partial} u = f$. Finally, (6) is precisely (1). This proves Theorem 1.

5. Why the condition must couple components

A componentwise copy of the one-variable moment condition cannot work. Indeed, choose $a \in C_c^\infty(\mathbb{C}^n)$ with $\bar{\partial}_2 \bar{\partial}_1 a \not\equiv 0$ and set $f = (\bar{\partial}_1 a, 0, \dots, 0)$. Each component annihilates all entire scalar tests by integration by parts, yet $\bar{\partial} f \neq 0$, so no scalar u can have $\bar{\partial} u = f$. The divergence-free vectors in (C) are the exact coupled replacement: they describe the full adjoint kernel and implicitly encode the usual $\bar{\partial}$ -compatibility.

When $n = 1$, $\mathcal{Z}_\phi = A^2(\mathbb{C}, e^{-2\phi})$, because $\bar{\partial} g = 0$ distributionally means that g is entire. Also $L_\phi = \Delta \phi$ in the source's normalized convention. Thus both the hypothesis and the constant reduce verbatim to Theorem 1.2.

Upgrade and novelty audit. Eight focused stages were completed: exact-signal and formulation audits; rejection of componentwise moments; conjugated-operator reformulation; trace-Levi identity; adjoint-kernel identification; closure construction; and reduction, constant, and literature checks. Bounded primary-source searches through 17 August 2026 found no arXiv paper stating this exact growing-weight result. The Hilbert-space mechanism is elementary enough that the result may be folklore; priority requires expert review.

References

- [1] H. Hedenmalm, *On Hörmander's solution of the $\bar{\partial}$ -equation*, arXiv:1311.2020 (2013).
- [2] L. Hörmander, *An Introduction to Complex Analysis in Several Variables*, 3rd ed., North-Holland, 1990.